

https://github.com/SETAPESU26/02_air_hockey


I'll start by looking at the repo to see what's broken and what features the README asks for.

The README is clear. Next I'll pull the actual code so I can find the bug.

GitHub's page blocks automated fetching, but my sandbox can reach GitHub directly, so I'll clone the repo.

Now a headless test of everything: collisions at many speeds and angles, goals, timer, draw and restart.

All tests pass. I'll package the result as the code folder plus one patch per task commit.

I cloned `SETAPESU26/02_air_hockey`, found the bug, and implemented all four tasks as separate commits. I tested them headlessly, not by playing the game. The tests covered 144 puck-into-paddle runs at 6 speeds and 24 angles, with no penetrations. They also covered goals, wall bounces, the timer edge (29.9s vs 30.0s), restart, and a 31-second random-input soak. You still need to run `python main.py` and do the videos yourself.

What was wrong and what changed

Task	Problem / change	File(s)
1. Collision	On contact the code only flipped <code>vx</code> . The puck was never pushed out of the paddle, so it vibrated inside it, and top or bottom hits bounced wrongly. It now reflects off the contact normal and gets pushed out. The puck moves in small sub-steps so fast shots can't tunnel through. Speed is capped, and a moving paddle transfers its velocity to the puck.	<code>collisions.py</code> , <code>paddle.py</code> , <code>game_engine.py</code>
2. Scoring	<code>player_score</code> and <code>computer_score</code> are shown on screen. A goal counts only when the puck is in the gap and fully past the end line. Hitting the wall elsewhere just bounces.	<code>game_engine.py</code>
3. Timer	A 30s countdown is shown. At zero the game freezes and shows "YOU WIN!", "COMPUTER WINS!" or "DRAW". R restarts, which the README promised but <code>main.py</code> never handled.	<code>game_engine.py</code>

Tas k	Problem / change	File(s)
4. R ese t	The original reset set velocity to 0, so the puck sat dead. It now returns to center and serves immediately toward whoever conceded. Paddles and AI state reset too.	game_eng ine.py

Getting it into your repo

The zip has an `air-hockey` folder inside `Lab-4/`, ready to drop into your own repo. There is also one `.patch` per task, so you can keep separate commits:

bash

```
git clone https://github.com/<you>/<your-lab-repo>.git && cd <your-lab-repo>
mkdir Lab-4 && cd Lab-4
# copy the ORIGINAL air-hockey folder in, commit it, then:
git am /path/to/patches/000*.patch # applies 4 separate commits
git push
```

Don't open a PR to SETAPESU26.

Things to be aware of

- **Chat history:** the lab requires your own exported chat history showing 3–4 attempts per task. A one-shot paste of my output won't show the iterative prompting. Walk through the tasks one at a time in your own chat, and use my code as a reference to review against.
- **Suggested prompts:** for the bug, try something like *"Here's collisions.py. The puck sticks inside the paddle and bounces wrong when hit from above. Fix it using the contact normal and resolve penetration."* For the features, one prompt each.
- **Videos:** record 10s before changes (the puck jitters inside a paddle and stalls after a goal) and 10s after. Press R at any time to restart so you can show a goal, the timer and the end banner within 10 seconds.
- **Submission:** the repo README lists the chat link as a submission item, while the handout asks for a doc/PDF export. Include both.

If you want, I can write short per-task prompts for your chat, or adjust the serve behavior or timer position.

read the instructions given in the lab pdf completely and check if everything done is proper or not



I'll re-check each item in the handout against what I actually delivered, and test the parts I haven't verified yet: each commit on its own, the puck pinned against a wall, the on-screen layout, and the push workflow I described.

Now I'll fold the fix into the Task 1 commit, then re-run the pinned-puck tests and the full suite.

Pinned-puck cases now resolve with zero penetration. Next I'll look at the actual rendered frames, since I haven't seen the HUD yet.

Both screens look right. Last, I'll regenerate the patches and test the exact push workflow on a simulated "Lab repo". My earlier instructions (running `git am` inside `Lab-4`) were untested.

I rechecked everything against the handout. The first version had two real bugs and one wrong instruction, and all three are now fixed and re-tested. The fixed code and patches are in the cards above.

What I found and fixed

Issue	Why it mattered	Status
Puck stuck when pinned against a wall or corner	The README says the puck must "never get stuck vibrating inside a paddle". My first version still did this when a paddle held the puck against a wall (it stayed 16px from the paddle's center, inside the 40px contact distance).	Fixed. The puck now slides out along the wall. Six pin scenarios show 0.00px penetration.
Commits authored as Student <student@example.com>	Your repo history would show a fake identity.	Re-author step added (see below).
My <code>git am</code> instructions didn't work	I tested them: running <code>git am</code> inside <code>Lab-4/</code> fails with "does not exist in index".	Corrected below.

Handout checklist

Handout item	Status
Fix the broken code (Task 1)	Done
Add all README features (scoring, 30s timer, reset)	Done
Separate commit per task	Done: 4 commits, and each one runs on its own
Code runs	Tested headlessly: collision sweep (0 penetrations), 8 full 30s random-input matches, and screenshots of the scores, timer and end banner. I haven't played it with a real window.
Don't raise a PR to SE TAPESU26	Nothing to do on my side
Videos (before and after, 10s each)	You must record these
Chat history exported as doc/pdf	You must export this
Push to your own repo under Lab-4/	You must do this

Correct push steps

Run these from the root of your own lab repo:

```
bash
```

```
mkdir Lab-4
# copy the ORIGINAL starter (air-hockey/, .gitignore, README.md) into Lab-4/, then:
git add . && git commit -m "Add original starter"

git am --directory=Lab-4 /path/to/patches/*.patch          # 4 separate commits
git rebase --exec 'git commit --amend --reset-author --no-edit' HEAD~4  # your name on them
```

Then copy your videos and exported chat into Lab-4/ and push. I tested this exact sequence on a simulated repo. The result matched my tested code byte for byte.

To run the game, `cd Lab-4/air-hockey` , then `pip install -r requirements.txt` and `python main.py` . The code must be run from that folder.

Things I can't verify for you

- **Individual work and prompting.** The handout says "individual assignment" and wants the job done in 3–4 prompts per task, with your chat history as a deliverable. I wrote all four tasks in one go, so my output on its own won't show your prompting. The sensible way to use it is to work through the tasks one at a time in your own chat, then compare with my code. Make sure you can explain the collision fix, since you may be asked about it.
- **Submission format.** The handout asks for the chat as a doc/pdf in your repo, while the repo's README checklist asks for a chat link. Submit both to be safe.
- **Video tip.** Press R to restart, so you can show a goal, the timer and the end banner within 10 seconds.